

14.0 POWER-DOWN

The ENC28J60 may be commanded to power-down via the SPI interface. When powered down, it will no longer be able to transmit and receive any packets.

To maximize power savings:

1. Turn off packet reception by clearing ECON1.RXEN.
2. Wait for any in-progress packets to finish being received by polling ESTAT.RXBUSY. This bit should be clear before proceeding.
3. Wait for any current transmissions to end by confirming ECON1.TXRTS is clear.
4. Set ECON2.VRPS (if not already set).
5. Enter Sleep by setting ECON2.PWRSV. All MAC, MII and PHY registers become inaccessible as a result. Setting PWRSV also clears ESTAT.CLKRDY automatically.

In Sleep mode, all registers and buffer memory will maintain their states. The ETH registers and buffer memory will still be accessible by the host controller. Additionally, the clock driver will continue to operate. The CLKOUT function will be unaffected (see **Section 2.3 “CLKOUT Pin”**).

When normal operation is desired, the host controller must perform a slightly modified procedure:

1. Wake-up by clearing ECON2.PWRSV.
2. Wait at least 300 μ s for the PHY to stabilize. To accomplish the delay, the host controller may poll ESTAT.CLKRDY and wait for it to become set.
3. Restore receive capability by setting ECON1.RXEN.

After leaving Sleep mode, there is a delay of many milliseconds before a new link is established (assuming an appropriate link partner is present). The host controller may wish to wait until the link is established before attempting to transmit any packets. The link status can be determined by polling the PHSTAT2.LSTAT bit. Alternatively, the link change interrupt may be used if it is enabled. See **Section 12.1.5 “Link Change Interrupt Flag (LINKIF)”** for additional details.

TABLE 14-1: SUMMARY OF REGISTERS USED WITH POWER-DOWN

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
ESTAT	INT	BUFER	r	LATECOL	—	RXBUSY	TXABRT	CLKRDY ⁽¹⁾	13
ECON2	AUTOINC	PKTDEC	PWRSV	r	VRPS	—	—	—	13
ECON1	TXRST	RXRST	DMAST	CSUMEN	TXRTS	RXEN	BSEL1	BSEL0	13

Legend: — = unimplemented, read as '0', r = reserved bit. Shaded cells are not used for power-down.

Note 1: CLKRDY resets to '0' on Power-on Reset but is unaffected on all other Resets.

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